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ROLL No : 22CSB07
DEPT : CSE
SUBJECT : Object Oriented Programming (OOPS)

PART A

1. Object Oriented Programming is programming using objects. It is called OOPS It uses class and object.

2 Encapsulation means encapsulation only. It means data hiding We hide data inside class.

3. Inheritance means inheriting.
Child class takes properties from parent class.

4. Polymorphism means "many forms". It allows the same function or method to behave differently in different situations.

5. Class is a blueprint or template used to create objects. Example: Car is a class.
Object is an instance of a class. Example: BMW is an object.

PART B

6 Principles of OOPS are

* Encapsulation
* Inheritance
* Polymorphism
* Abstraction

a) Encapsulation means hiding data
b) Inheritance means child class
c) Polymorphism means many forms
d) Abstraction means abstract class

7 Types of inheritance

* Single inheritance
* Multiple inheritance
* Multilevel inheritance
* Hierarchical inheritance.

Single : A → B
Multiple : A → B → C , D
Multilevel : A → B → C
Hierarchical : A → B , A → C

8 Constructor is constructor and it is called when object is created.
Destructor is destructor and it is called when object is destroyed

Example

```
class A {
public:
  A() { }
~A() { }
};
```

9 * Polymorphism means many forms
* Compile time polymorphism means at compile time.
* Run time polymorphism means at run time
* Compile time example: function overloading
* Run Time example: Overriding

10 Exception handling
* Exception handling is handling exception
* It is used to handle errors.
* It uses try and catch

Example

```
Try {
  int a = 10/0,
}
catch ( ) {
}
```